



PixelProbe

PIXELPROBE – AI IMAGE DETECTION SYSTEM

NAOISE O'LOUGHLIN

SOFTWARE & ELECTRONIC ENGINEERING

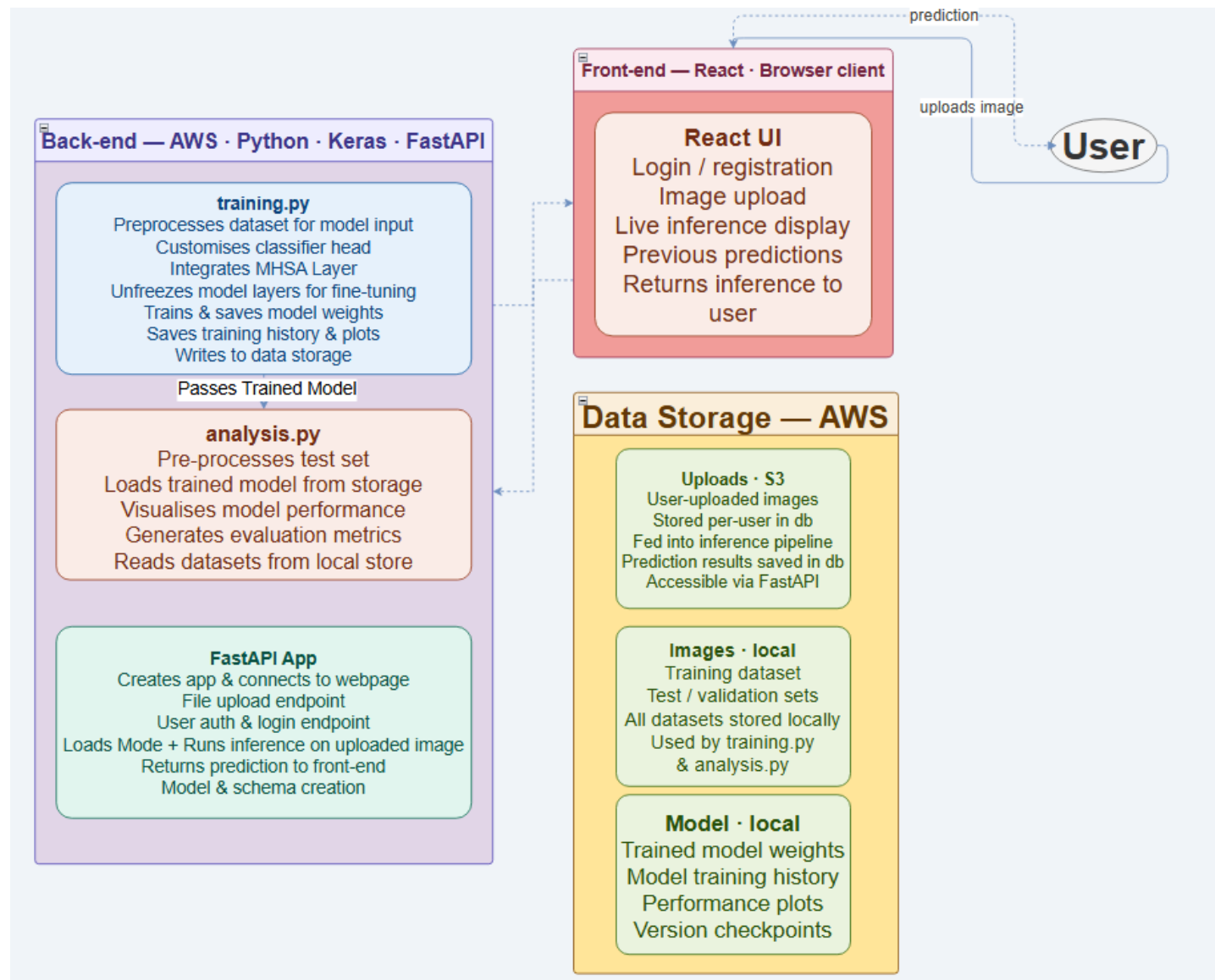
ATLANTIC TECHNOLOGICAL UNIVERSITY, GALWAY

INTRO

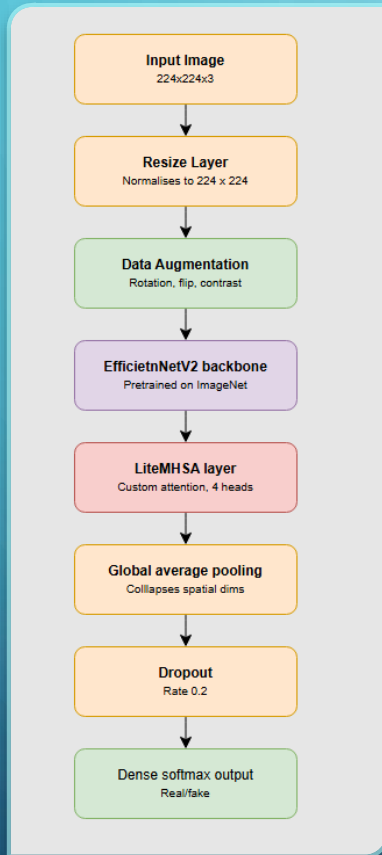
- Huge advancements in image generators like Dall-E and Midjourney over the past 3 years.
- Increasingly Difficult to distinguish between real and synthetic media.
- PixelProbe allows users to upload image and receive real time classification of real or fake images.
- Huge needs with the increase in miss-use of image generators such as the Grok controversy in December 2025.

SYSTEM ARCHITECTURE

- React Frontend provides a reactive user interface
- FastAPI backend handles model loading, predictions, user authentication. Hosted on AWS.
- User information & Prediction information stored in SQLite database.
- Image uploads stored in amazon S3.



MODEL PIPELINE



- **Input image:** 224 x 224 x 3
- **Resize Layer:** Resize + normalization for consistent model behavior.
- **Data Augmentation:** Introduces Random augmentations such as rotations, contrast change etc. to help improve generalisation.
- **EfficientNetV2 Backbone:** Pretrained on ImageNet, provides hierarchal feature representations
- **LiteMHSA:** Multi-Head Self Attention Layer that aids in detection of subtle, fine-grained artifacts
- **Global Average Pooling:** Compresses spatial features
- **Dropout:** 0.2 for regularisation
- **Dense softmax output:** Outputs final class probabilities (real vs fake)

LITE MHSA

Purpose

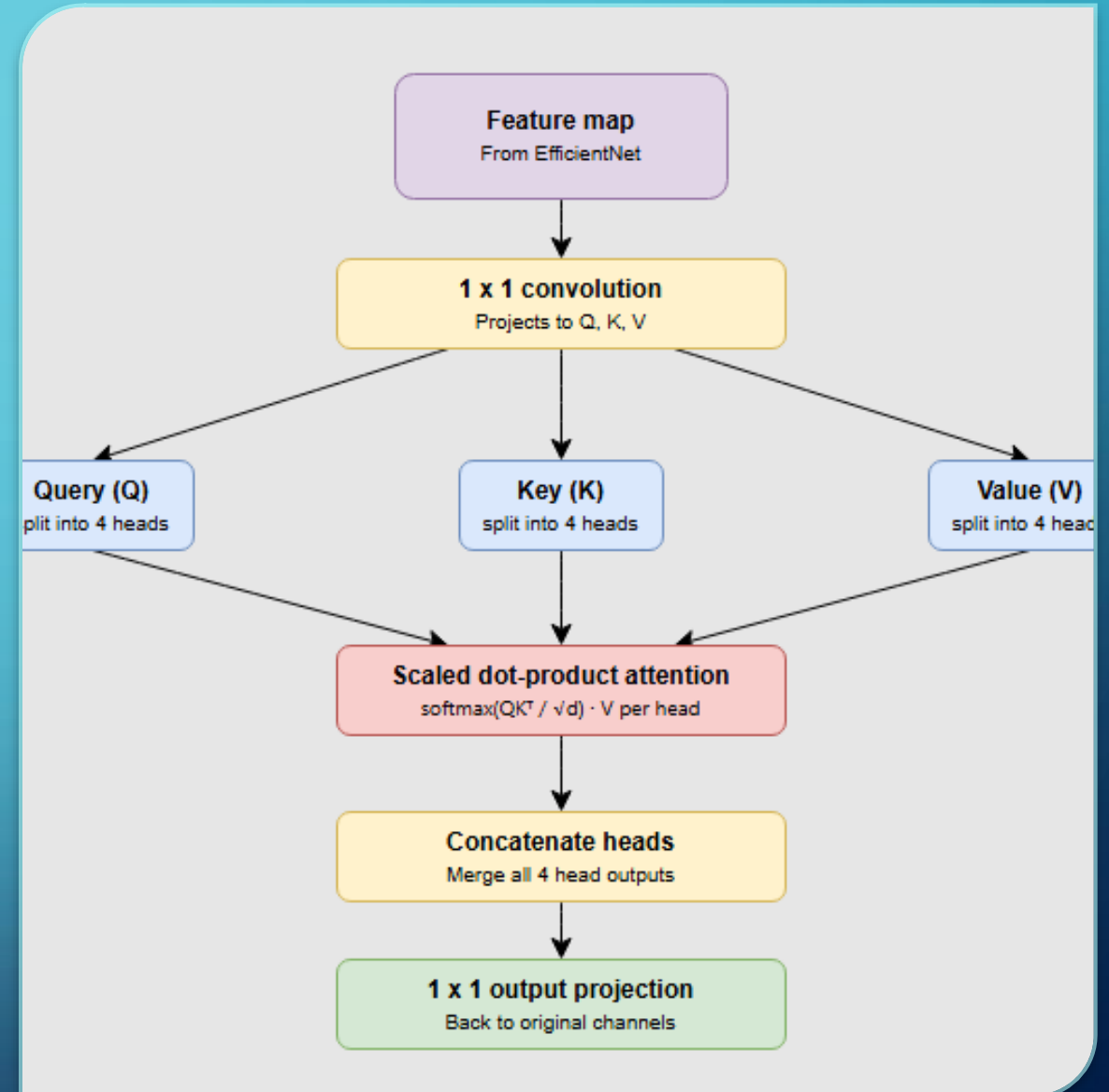
- Adds global context so the model can detect subtle, long-range artifacts that CNNs miss.

How It Works

- **1×1 Conv** → **Q, K, V**: Projects features into attention space
- **4 Heads**: Each head focuses on different patterns
- **Scaled Dot-Product Attention**: Computes similarity across all spatial positions
- **Concat + 1×1 Output**: Merges heads and restores channel depth

Why It Helps

- Captures **non-local relationships**
- Improves detection of **fine-grained inconsistencies**
- Lightweight and **computationally efficient**



Mode	Train Set	ZHANG	CIFAKE	GEN/P365	GHA1
PixelProbeB0	ZHANG	86	50	58	70
PixelProbeB1 V3	ZHANG	94	51	57	<u>78</u>
PixelProbeB1CFV2	CIFAKE	44	95	51	26
PixelProbeB1GI	GenImage	52	<u>61</u>	80	77
PixelProbeB2GI	GenImage	50	56	<u>79</u>	75
PixelProbeB2GIV3	GenImage/Places365	<u>56</u>	54	78	73

RESULTS

IMAGE CLASSIFICATION CROSS-DATASET ACCURACY RESULTS (%)

CONCLUSION



All planned features fully implemented and validated across the system.



Provides reliable detection of AI-generated content in real-world scenarios.



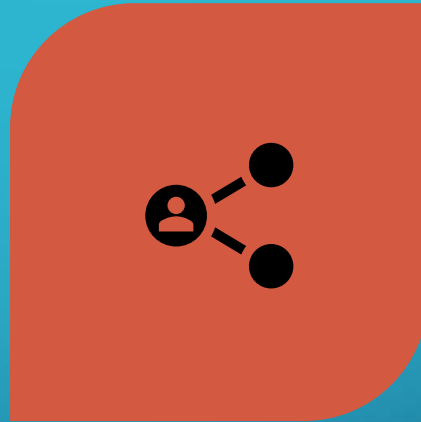
PixelProbe is now a complete and scalable AI-Image Detection system.

FUTURE WORK



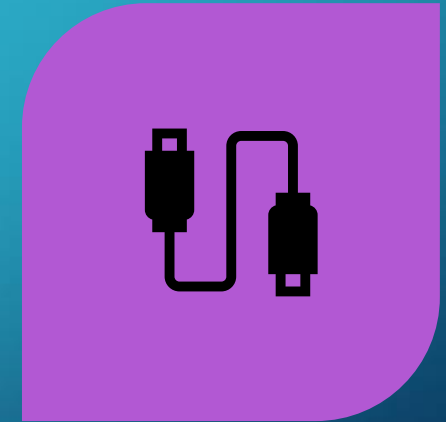
VIDEO FRAME-BY-FRAME DETECTION

INTEGRATE REAL-TIME FRAME-BY-FRAME DETECTION SHOWING USERS SECTIONS OF VIDEO THAT MAY BE TAMPERED WITH.



MOBILE APP DEPLOYMENT

A MOBILE APP WOULD MAKE PIXELPROBE MORE ACCESSIBLE TO USERS AND EASIER WAY TO VERIFY IMAGES ON SOCIAL MEDIA.



PLUGINS

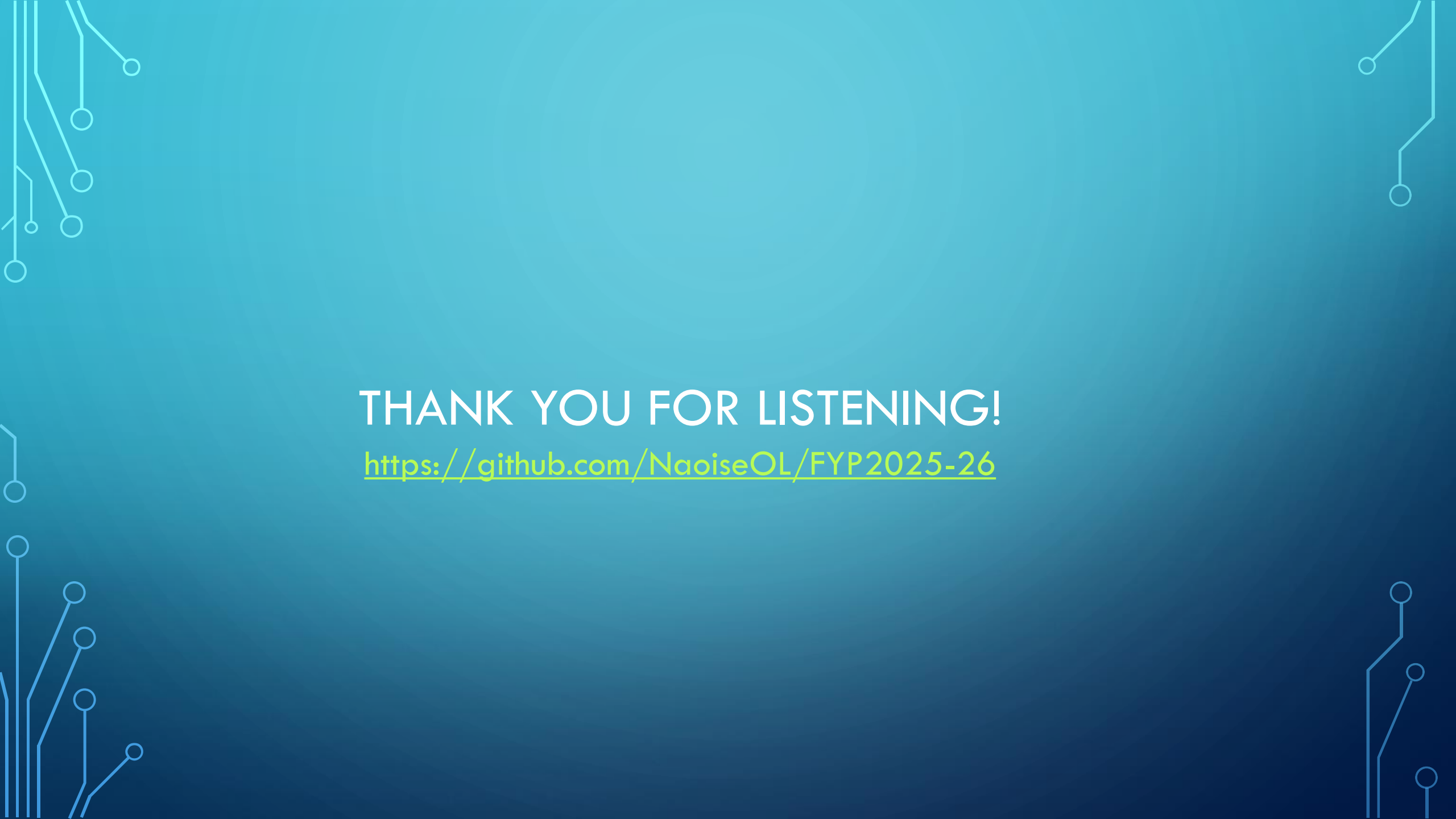
INTRODUCING PIXELPROBE AS A WEB OR MOBILE PLUGIN WOULD REMOVE THE NEED FOR USERS TO DOWNLOAD IMAGES TO THEIR DEVICE FOR INFERENCE.

AI USAGE

- Over the course of the project, AI assistants Claude and Copilot have been used.
- The use of the AI assistants have been for research purposes and Bug fixes predominantly.
- Additional usage has been to gain deeper understanding of source code referenced throughout the development of this system.

PROJECT TIMELINE

Work		September '25	October - December '25	January - March	April - June	July - September	
> FN-4 Project Proposal	DONE						
> FN-8 Preprocessing	DONE						
> FN-12 Initial Model Traini...	DONE						
> FN-14 Initial Model Analysis & Evaluati...	DONE						
> FN-34 Model Generalizati...							
> FN-13 Backend API							
> FN-16 Database							
> FN-17 Frontend React...							
> FN-52 AWS Integration							
> FN-22 Project Poster							
> FN-20 Report Write Up							
> FN-21 Presentation/Demo							

The background is a blue gradient with decorative white circuit-like lines in the corners. The lines consist of straight segments and small circles, resembling a stylized electronic circuit.

THANK YOU FOR LISTENING!

<https://github.com/NaoiseOL/FYP2025-26>